

WASP-43b

R-1877

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_190313 · created 2026-08-03T11:24:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458555.803 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.058
Transit depth (Rp/Rs)^2	3.78 +/- 2.26 [%]
Orbital Inclination (inc)	85.37 +/- 2.87
Airmass coefficient 1 (a1)	2.94 +/- 1.77
Airmass coefficient 2 (a2)	-0.77 +/- 0.91
Scatter in the residuals of the lightcurve fit is	55.06 %
Best Comparison Star	#3 - [418, 141]
Optimal Aperture	1.93
Optimal Annulus	5.75
Transit Duration (day)	0.0589 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.058 vs 0.1594 (deviation 0.6σ)
Duration fit vs published	0.0589 d vs 0.0512 d (deviation 1.11σ)
Mid-time O–C	-3.9 min
Depth SNR	3.3
Residual scatter	55.06 %

Light curve

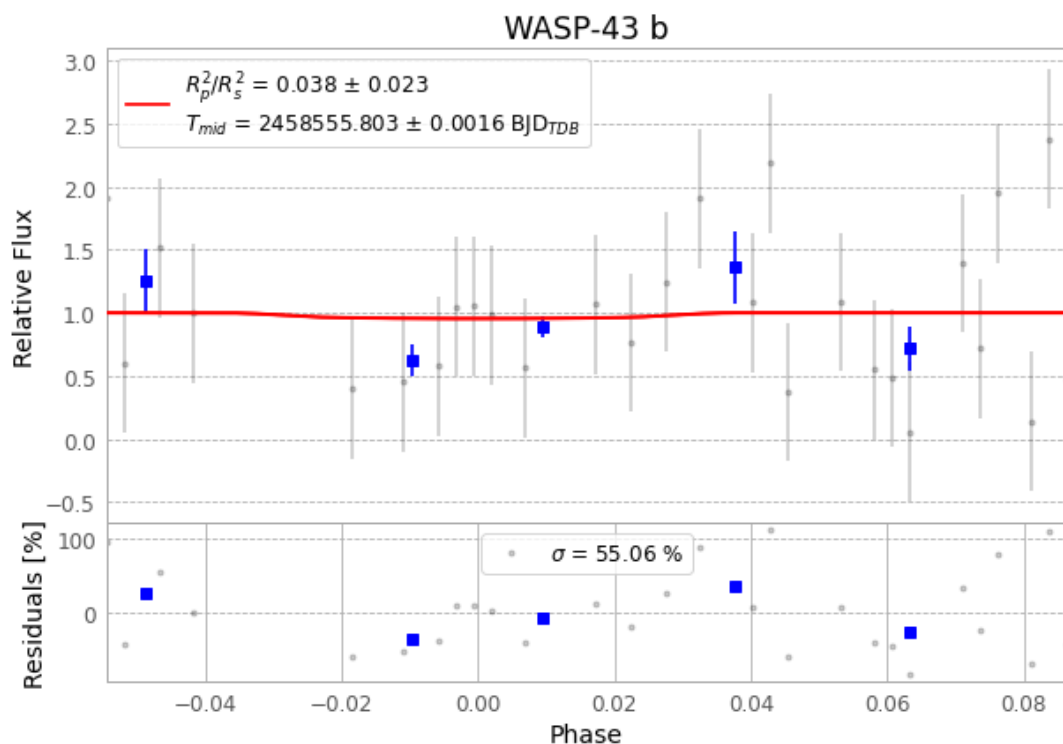


Figure 1 — FinalLightCurve_WASP-43 b_2019-03-12.png (SHA-256 994a0f76d8313688...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [135, 165], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8d0146fe6662a369...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), WASP-43190313055412.FITS → WASP-43190313090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190313.log (ea7ae9d5261d...), FinalLightCurve_WASP-43 b_2019-03-12.png (994a0f76d831...), FinalLightCurve_WASP-43 b_2019-03-12.csv (cc6e9f93dff6...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:24Z.